

AI TOOLS FOR RESEARCH

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Background

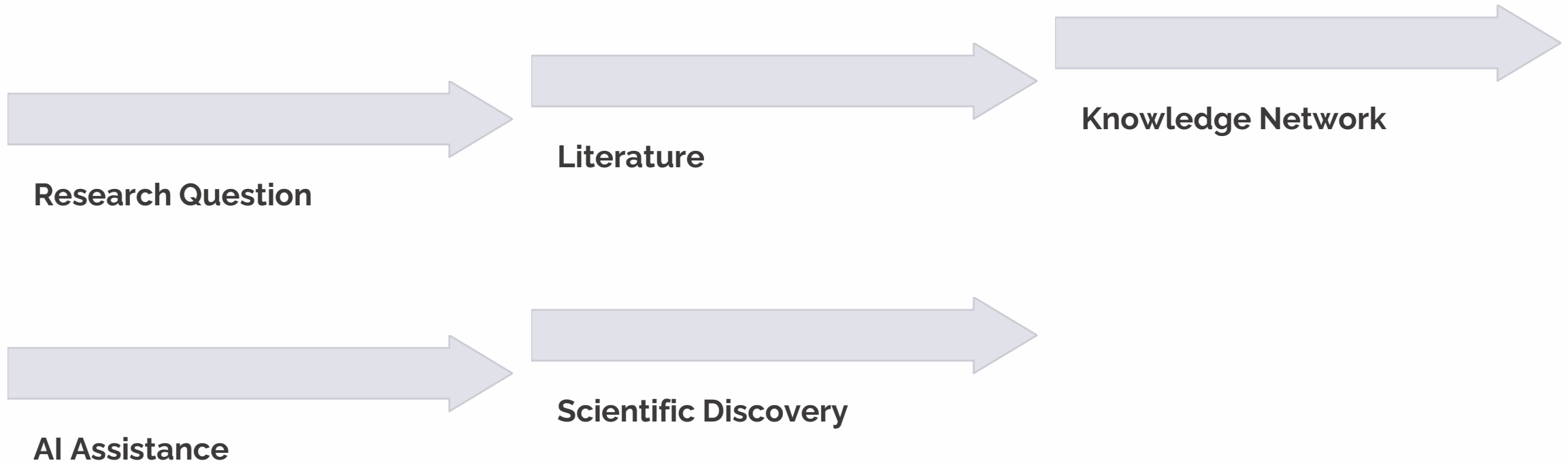
Next: How Do We Talk to AI Effectively?

Prompt Engineering and Evaluation of AI Research Tools



“A powerful AI tool with a poor research question is still a poor research process.”

AI-Assisted Research: From Search to Scientific Discovery



If AI Can Search, Summarize, Analyse and Write... What Does the Researcher Do?



What Remains Uniquely Human? → Scientific Judgment

The capacity to evaluate evidence, weigh uncertainty, and take intellectual responsibility for conclusions.

The Modern Researcher's Challenge



Publication Explosion

2M+ new papers per year



Interdisciplinary Research

Relevant work scattered across fields



Rapidly Changing Topics

Methods shift within months



Citation Overload

Thousands of references per topic



Difficulty Identifying Novelty

Hard to know what is truly new

Two Ways of Navigating the Literature

Traditional Workflow

01

Keyword Search

02

Read Papers

03

Take Notes

04

Compare Studies

05

Identify Gap

AI-Assisted Workflow

01

Semantic Search

02

Citation Exploration

03

AI-Assisted Screening

04

Knowledge Mapping

05

Evidence Synthesis

06

Human Validation

"AI increases speed and scale."

The additional step — **Human Validation** — is not optional; it is what makes AI-assisted research defensible.

AI-Assisted Research ≠ AI-Generated Research

AI-Assisted Research ✓

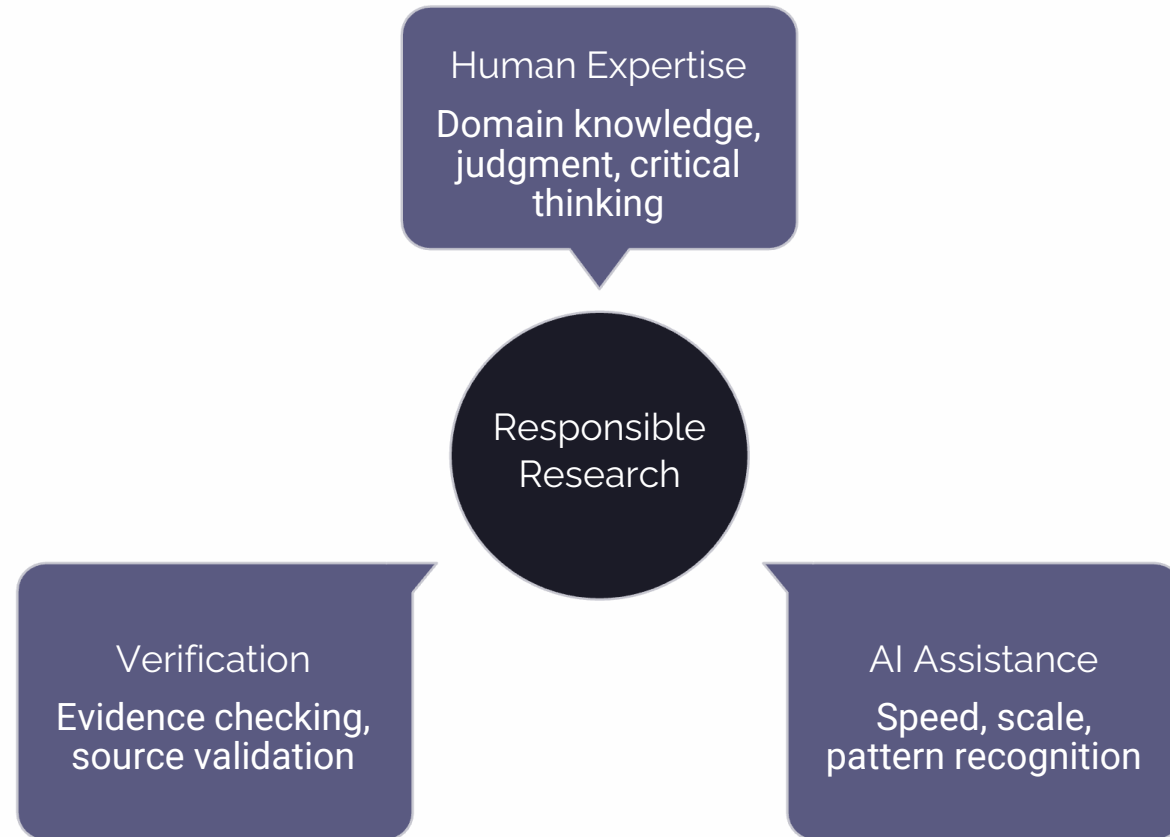
- Human defines the problem
- AI accelerates tasks
- Evidence is verified
- Researcher interprets findings
- Researcher owns decisions

AI-Generated Research ⚠

- AI drives conclusions
- Sources may be unchecked
- Reasoning may be opaque
- Hallucination risk increases
- Accountability becomes unclear

"Use AI to support reasoning, not replace responsibility."

A Responsible Model of AI-Assisted Research



Human Expertise × AI Assistance × Verification = Responsible Research

How Scientific Research Has Evolved



Library-Based Research

Manual catalogues, physical journals



Digital Research

Online databases, keyword search



Data-Driven Research

Large datasets, statistical computing



AI-Assisted Research

Semantic search, AI reading,
knowledge graphs



Agentic Research

Autonomous AI agents executing
research tasks

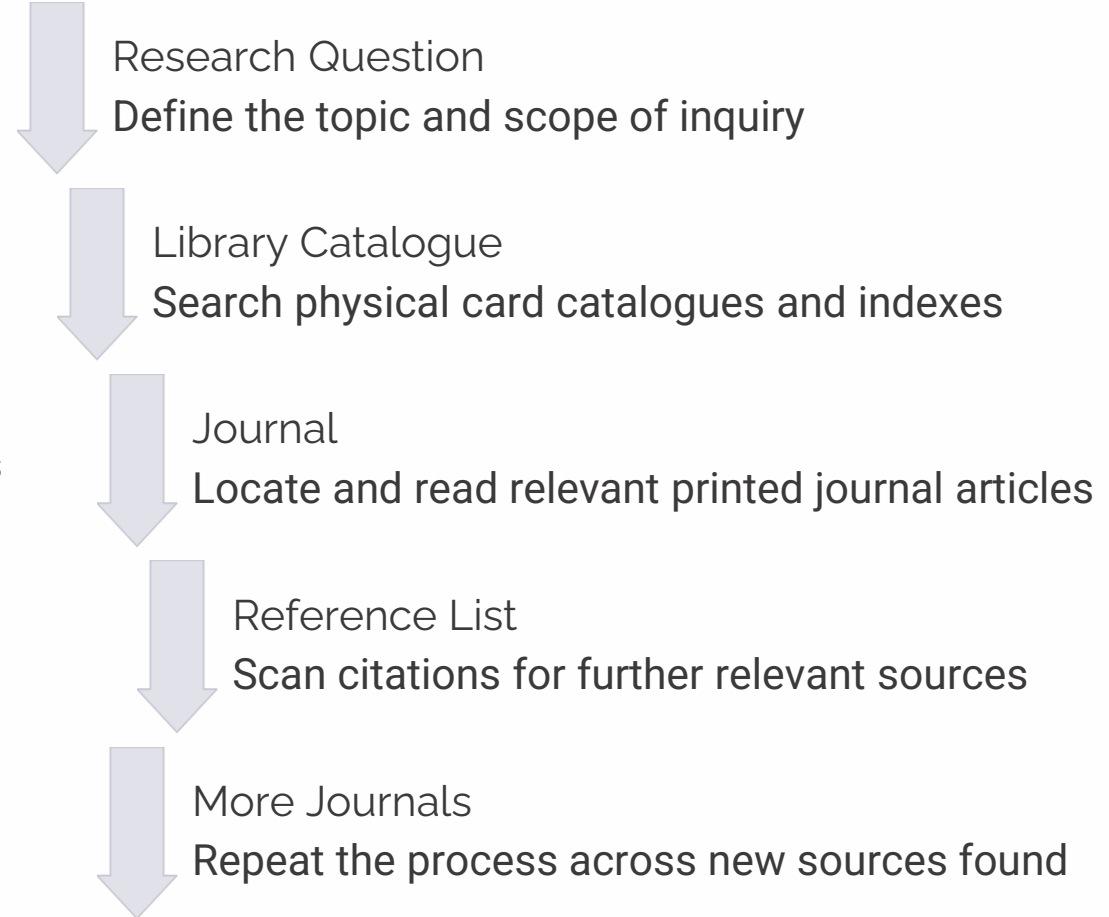
📄 The **AI-Assisted** and **Agentic** stages represent the current frontier. This lecture focuses on AI-Assisted research – where human oversight remains central.

Research Era 1: Library-Based Discovery

Primary Sources

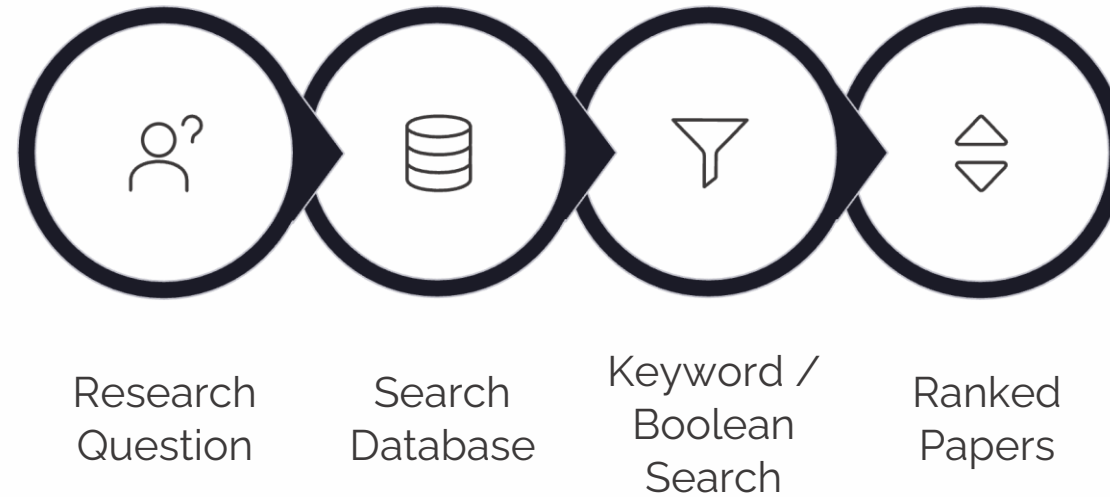
-  Books
Monographs and textbooks as primary reference
-  Printed Journals
Peer-reviewed articles distributed by subscription
-  Printed Indexes
Manual bibliographies and subject catalogues
-  Manual Bibliographies
Reference lists compiled by hand from cited works

Discovery Workflow



*Discovery depended heavily on manual reference tracing.
Access to knowledge was constrained by geography,
institutional subscriptions, and personal networks.*

Research Era 2: Digital Scholarly Search



Google Scholar

Broad academic web indexing across disciplines and grey literature



Web of Science

Citation indexing and impact metrics



Scopus

Peer-reviewed journal coverage with citation metrics and author profiles

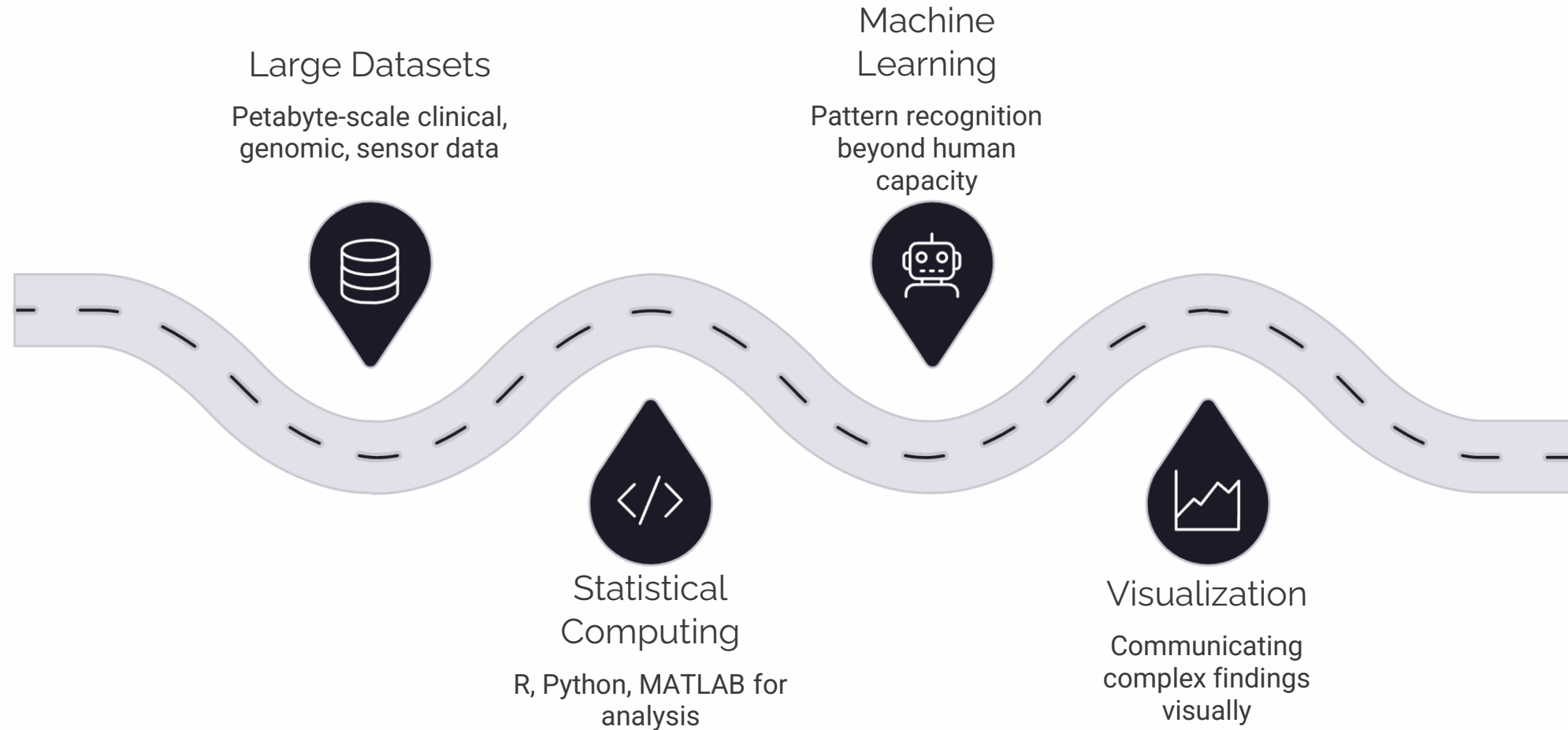


PubMed

Biomedical and life sciences — MEDLINE indexed, free access

| *Search became faster, but researchers still needed to know what to search for.*

Research Era 3: Data-Driven Science

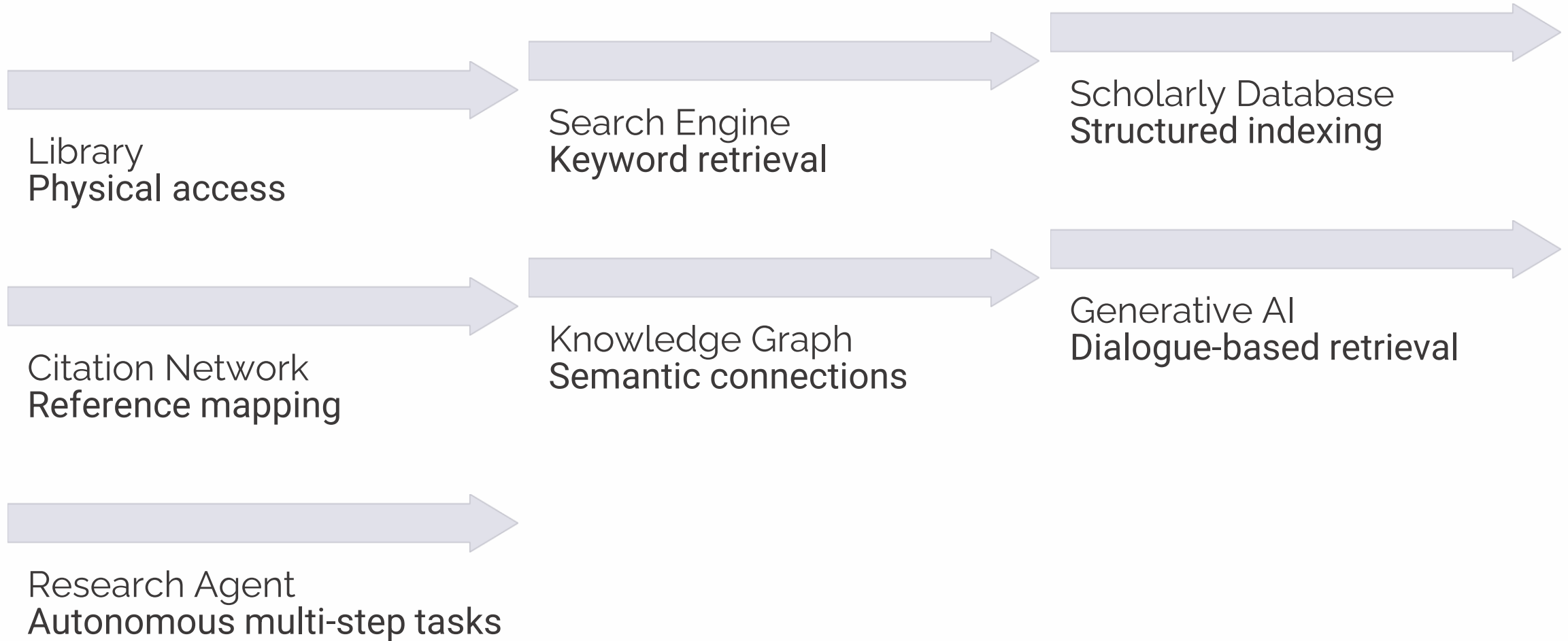


Research Era 4: AI-Assisted Science

"AI is becoming embedded across the research workflow."

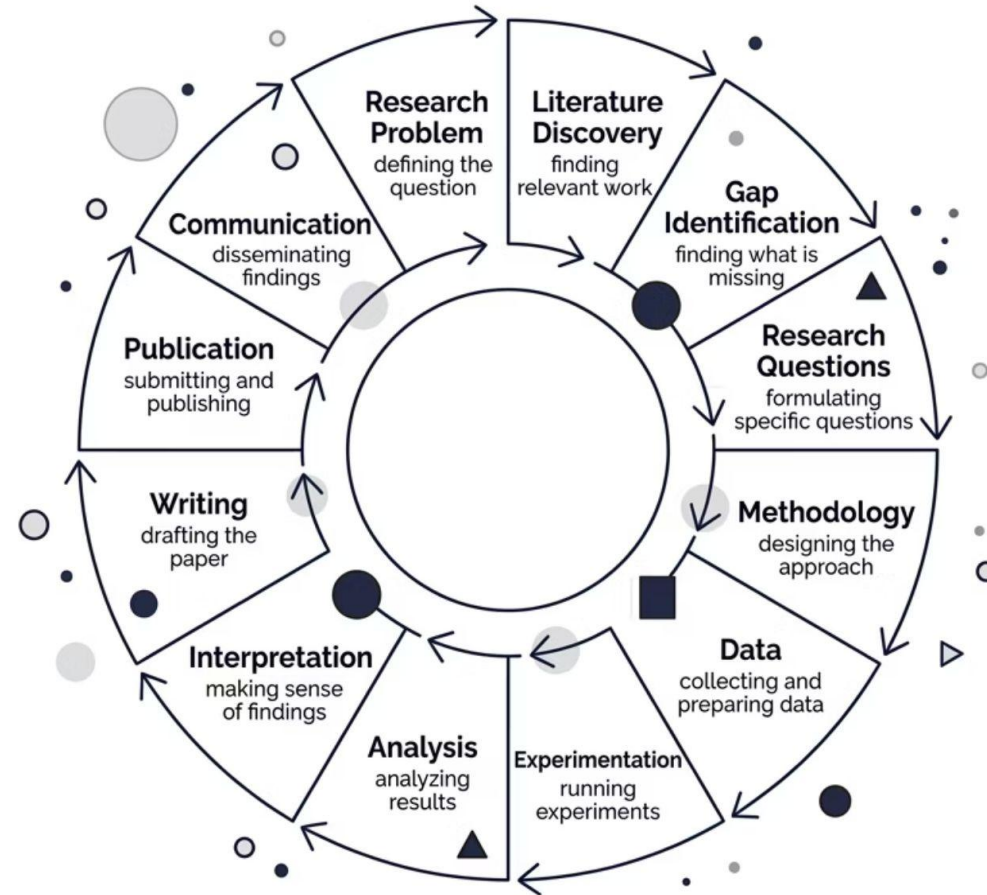
No single stage remains untouched — but human oversight remains essential at every stage.

From Search Engines to Research Agents



| **The interaction model is moving from "search" to "dialogue + action".**

The Complete Research Lifecycle



Each stage presents specific opportunities for AI assistance — and specific responsibilities that remain with the researcher.

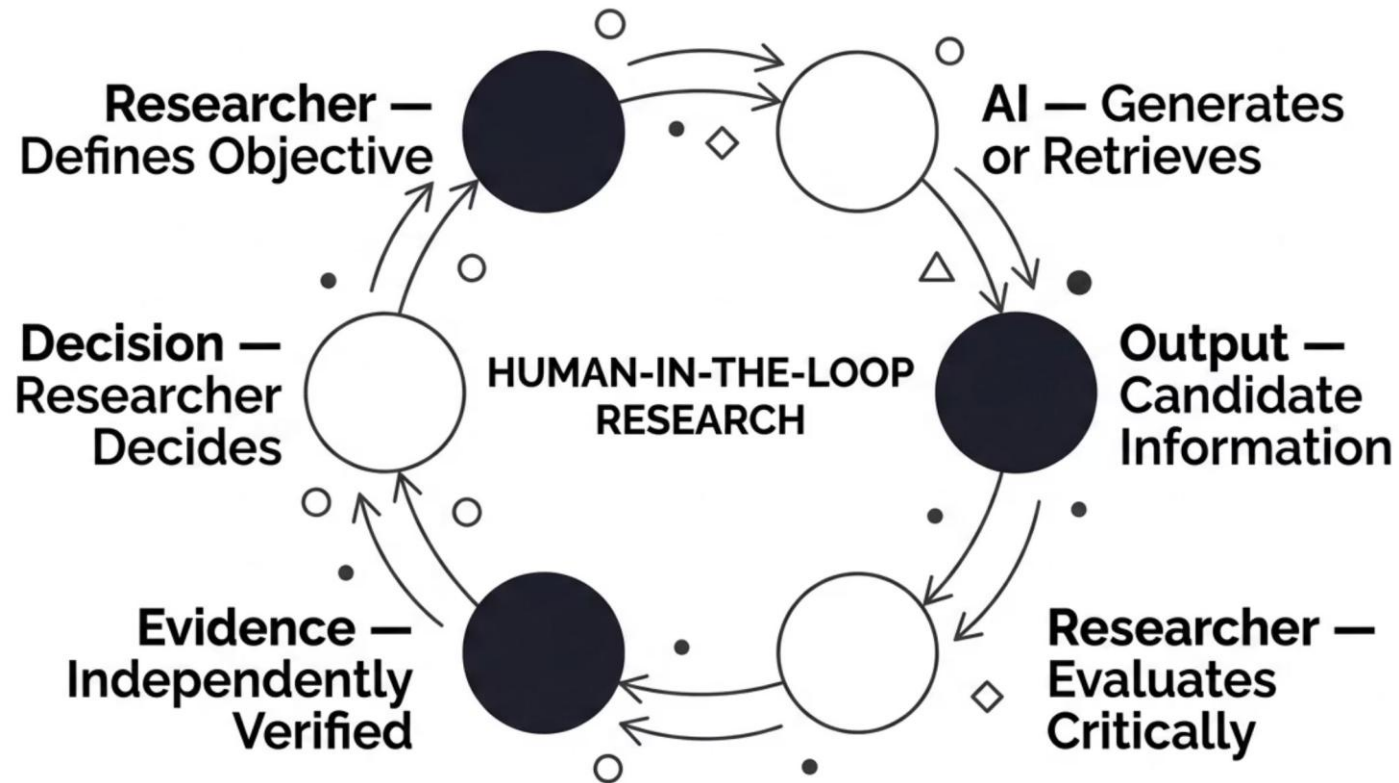
The AI-Assisted Research Responsibility Matrix

MAJOR ANCHOR SLIDE

Research Stage	AI Can Assist	Researcher Must Own
Problem	Exploration — terminology, related domains, alternative framings	Final definition — the specific, bounded research question
Literature	Discovery — semantic search, citation tracing, network mapping	Critical evaluation — relevance, quality, and validity of sources
Methodology	Comparison — surfacing alternative methods from the literature	Scientific validity — fitness for purpose and disciplinary standards
Analysis	Computation — code generation, statistical testing, pattern detection	Interpretation — what the results mean and why they matter
Writing	Language — clarity, grammar, structure, and style compliance	Claims — every factual and scientific assertion in the paper
Publication	Formatting — journal templates, reference formatting, figure preparation	Accountability — the researcher's name is on the paper

Assistance increases. Responsibility does not decrease.

Who Should Lead the Research: Human or AI?



The researcher leads every critical decision — AI assists but never decides.

AI Output Should Be an Intermediate Research Object

The Five-Step Human-in-the-Loop Workflow



RESEARCHER — Defines Objective
Sets the research question and scope



AI — Generates or Retrieves
Produces candidate information at scale



RESEARCHER — Evaluates
Critically assesses AI output for quality and relevance



EVIDENCE — Verifies
Claims checked against primary, citable sources



RESEARCHER — Decides
Makes the final, accountable research decision

AI output is a starting point, not a conclusion.

AI output is not a research finding. It is a **candidate** — a step in the process that must be evaluated, verified, and decided upon by the researcher before it becomes evidence.



Candidate

Plausible but unverified — requires researcher judgment before use



Intermediate

Requires researcher evaluation — never the final conclusion



Verified

Validated and citable — becomes true research evidence

What Humans Still Do Better



Scientific Judgment

Evaluating whether evidence is sufficient and valid



Contextual Understanding

Interpreting findings within disciplinary norms



Creativity

Generating genuinely novel hypotheses



Causal Reasoning

Distinguishing correlation from causation



Ethical Responsibility

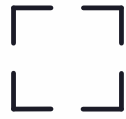
Owning the consequences of research decisions

What AI Does Better



Speed

Processing millions of documents in seconds



Scale

Handling corpora far beyond human reading capacity



Search Assistance

Semantic and citation-aware retrieval



Pattern Extraction

Identifying trends across large datasets



Repetitive Processing

Consistent execution of structured tasks

Human + AI = Complementary Strengths. The goal is not to choose between human and AI — it is to combine them intelligently.

Where AI Can Mislead Researchers

⚠️ Hallucination

Confident but factually false information — presented in fluent, authoritative language that makes errors difficult to detect

⚠️ Fabricated Citations

References that do not exist in any database — complete with plausible author names, journal titles, and DOIs

⚠️ Missing Context

Incomplete or decontextualized interpretation — accurate in isolation but misleading without the full picture

⚠️ Automation Bias

Humans trusting machine output without verification — the tendency to defer to AI because it sounds confident

⚠️ Overconfidence

Uncertainty hidden behind fluent, authoritative language — AI rarely says "I don't know" when it should

"Here are five highly cited papers on concept drift in healthcare..."

⊗ **Do these papers actually exist?** Always verify AI-generated citations against Scopus, Web of Science, or PubMed before citing them in your work.

AI Literacy for Researchers



UNDERSTAND

What is the tool doing? What model, training data, and method underlies it?



USE

Can I apply it appropriately to my research task?



EVALUATE

Is the output reliable, complete, and unbiased?



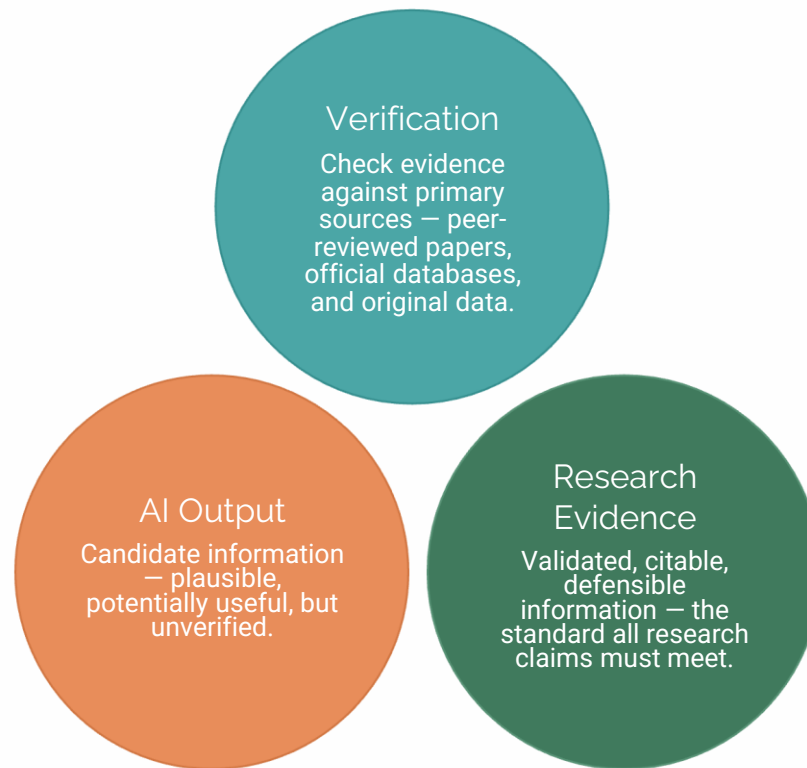
VERIFY

Can I independently confirm the output against primary sources?

"Using AI is not the same as being AI literate."

The Golden Rule of AI-Assisted Research

Trust, But Verify



Prompt Engineering Is Research Instruction Design



"A research prompt should specify both the task and the evidence standard."

What Is Wrong With This Research Prompt?

"Find papers on concept drift."

What would you add to make this a defensible research task?

Version 2: Add Research Scope

Weak Prompt

"Find papers on concept drift."

- No domain specified
- No time range specified
- No source type specified
- No output format specified
- No verification requirement

The AI will return whatever it finds most plausible — which may include fabricated references.

Scoped Prompt

"Identify peer-reviewed studies on concept drift detection in healthcare data streams published between 2021 and 2026."

Topic

Concept drift detection —
specific method class

Domain

Healthcare data streams —
specific application context

Publication Type

Peer-reviewed studies only —
evidence standard defined

Time Range

2021–2026 — recency
requirement specified

Specificity reduces ambiguity. Each added dimension narrows the search space and raises the evidence standard.

Version 3: Turn the Prompt Into a Research Task



Objective

What exactly do I want to know? Define the specific information need with precision.



Scope

What is included – domains, methods, populations, geographic contexts?



Evidence

What sources are acceptable – peer-reviewed journals, databases, recency thresholds?



Exclusions

What should be ignored – preprints, non-clinical settings, retracted papers?



Output

How should results be organized – comparison table, ranked list, structured synthesis?



Verification

How should uncertainty and gaps be handled – flag unknowns, request confidence levels?

Next: Build a complete research-grade prompting framework →

Research-Grade Prompt Engineering · AI Tool Evaluation · Scholarly Discovery · Citation Networks · Knowledge Mapping · Integrated Research Workflow

A Research-Grade Prompting Framework

R — C — S — E — O — V

R

Role

Who should the AI act as? Define expertise & analytical depth.

S

Scope

What are the limits? Define years, domain, study type & application.

O

Output

How structured? Specify format — table, synthesis, or comparison.

C

Context

What research problem? Provide domain, objective & prior knowledge.

E

Evidence

What sources qualify? Require title, authors, venue, year & DOI.

V

Verification

How handle uncertainty? Flag all unverifiable claims explicitly.

Good research prompts define both the task and the evidence standard.

R = Role: Define the Research Perspective

WEAK PROMPT X

"Find research."

No expertise signal. No depth expectation. No analytical frame. The AI defaults to generic, surface-level retrieval.

RESEARCH-GRADE PROMPT ✓

"You are assisting a PhD researcher conducting a structured literature exploration into concept drift detection in clinical prediction systems."

Specifying a role shifts the AI from general retrieval to **domain-calibrated analysis**.

1

Role Definition

Specify expertise, perspective & analytical depth

2

Analytical Depth

AI calibrates to domain-level rigor & standards

3

Research-Oriented Response

Targeted, evidence-grounded, structured output

i Role clarifies expertise — it is not a magic phrase.

C = Context: Tell the AI What Problem You Are Solving

Research Domain

Clinical informatics, machine learning in healthcare, predictive modelling.

Current Problem

Clinical prediction models deployed over long periods may experience **concept drift** – changes in data distributions and relationships.

Research Objective

Identify methods for detecting and managing concept drift in healthcare data streams.

Existing Knowledge

Drift detection is well-studied in general ML; healthcare-specific applications with **sparse labels remain underexplored**.

1

Context

Anchor the AI's interpretation with domain, problem & objective

2

Better Interpretation

AI aligns to your specific research frame and vocabulary

3

More Targeted AI Response

Evidence-grounded, domain-calibrated, structured output

"Context is the single most powerful component of a research prompt."

S = Scope: Define the Boundaries

INSIDE THE BOUNDARY ✓

Years: 2021–2026

Domain: Healthcare

Problem: Concept Drift

Study Type: Peer-Reviewed

Application: Clinical / Streaming Data

OUTSIDE THE BOUNDARY ✗

Blog posts, opinion pieces, non-scholarly sources

Unrelated domains (finance, NLP without clinical application)

Pre-2021 foundational-only work

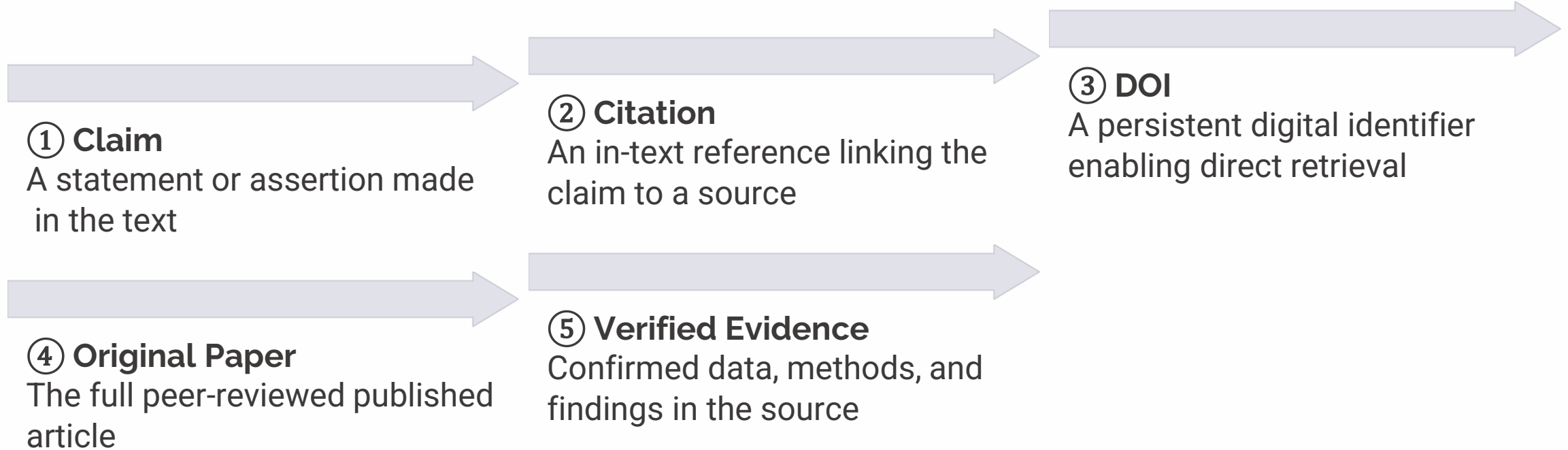
Preprints without peer-review status

Scope is a methodological decision — not a convenience setting.

A good research question also specifies what **NOT** to include.

E = Evidence: Ask Where the Claims Come From

Every scholarly claim must be traceable through a verifiable chain. Follow the evidence — step by step.



⚠️ **THE FLUENCY TRAP:** AI-generated text can be grammatically fluent, logically structured, and **factually wrong**. Fluency is a writing quality — **not** an evidence standard.

Fluent text is not scholarly evidence.

📌 **Prompt instruction:** "For each paper, provide title, authors, venue, year, and DOI. Do not invent references."

O = Output: Structure the Evidence for Comparison

Specifying output format transforms a list of references into a comparative research instrument.

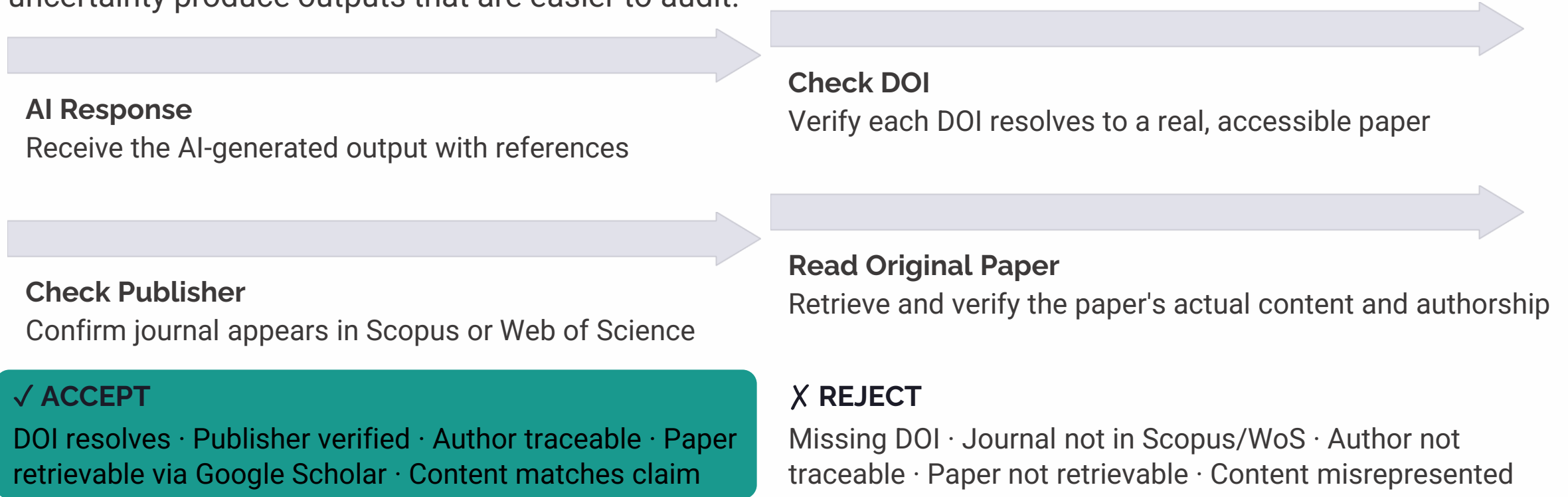
Paper	Problem	Method	Dataset	Key Finding	Limitation
Author et al. (2023)	Concept drift in EHR	ADWIN + LSTM	MIMIC-III	Detected drift 14 days earlier than baseline	Requires labeled data
Author et al. (2024)	Streaming clinical data	DDM + XGBoost	PhysioNet	Reduced false alarms by 31%	Single-site validation
Author et al. (2025)	Unsupervised drift	Autoencoder	Synthetic + real ICU	Works without labels	High compute cost



Structured output makes comparison easier.

V = Verification: Build Verification Into the Prompt

Verification should be **designed into the workflow**, not added at the end. Prompts that require the AI to flag uncertainty produce outputs that are easier to audit.



⚠ **Practical signals of unverified content:** Missing DOI · Journal not in Scopus/WoS · Author not traceable · Paper not retrievable via Google Scholar.

"Verification should be designed into the workflow, not added at the end."

From Casual Prompt to Research-Grade Prompt

CASUAL PROMPT ✗

"Find papers about concept drift."

What's missing:

No Role

No Context

No Scope

No Evidence Standard

No Output Format

No Verification

Prompt quality improves task quality.

Evidence verification determines research quality.

RESEARCH-GRADE PROMPT ✓

R — Role

"Act as a research assistant helping a **PhD student**..."

C — Context

"...investigate **concept drift detection** in **healthcare data**

S — Scope

"...peer-reviewed studies published between **2021–2026**. Prioritize clinical

E — Evidence

"...provide **title, authors, venue, year, DOI**, dataset, method, key result, and

O — Output

"...**Separate empirical studies from surveys**..."

V — Verify

"...**Do not invent references** and mark information that cannot be independently verified."

Three Questions to Ask About Any AI Research Tool

1. SOURCE BASE

Where does the information come from?

Is it a live scholarly database, web crawl, citation index, or LLM internal knowledge? Is the corpus transparent and documented?

2. MODEL CAPABILITY

What can the system actually do?

Search, retrieve, summarize, generate, analyse citations, map knowledge, detect trends? Does it conflate these capabilities?

3. RELIABILITY

Can the result be independently verified?

Are citations traceable? Is coverage broad enough for systematic review? Is hallucination risk disclosed?

Evaluate the research infrastructure, not only the interface.

1. Evaluate the Source Base

Prefer sources where the corpus is documented, the coverage is known, and individual claims can be traced to original papers.

LLM Internal Knowledge

No live retrieval. Trained on web-scale text up to a cutoff date. Cannot guarantee scholarly provenance. ⚠️ **High hallucination risk** for specific citations.

Web Search

Broad but unfiltered. Includes non-scholarly sources. Useful for grey literature but requires heavy filtering for academic use.

✓ Scholarly Database

Preferred. Curated, peer-reviewed content.

PubMed · IEEE Xplore · ACM DL

High precision. May have coverage gaps.

Citation Index

Tracks citation relationships across papers.
Enables impact analysis and network mapping.
Scopus · Web of Science

✓ Knowledge Graph

Preferred. Structured entities with explicit relationships.

OpenAlex · Semantic Scholar

Enables programmatic analysis and trend mapping.

What corpus is the answer grounded in?

2. Evaluate Model Capability

Understand what each capability actually does — and its risks — before trusting any tool for scholarly work.

Search

Retrieve papers from an indexed corpus. Not all tools search scholarly databases.

Retrieval

Surface specific papers or passages relevant to a query from a known source base.

Summarization

Condense paper content. **Risk:** may omit methodological nuance or misrepresent findings.

⚠️ Generation

HIGH hallucination risk. Produces new text — citations, statistics, or findings may be fabricated.

Citation Analysis

Map citation relationships, identify influential papers, and trace intellectual lineage.

Knowledge Mapping

Visualize conceptual clusters, research neighbourhoods, and field structure.

Trend Analysis

Track publication volume, topic emergence, and institutional activity over time.

"A tool that summarizes well may not discover literature comprehensively."

3. Evaluate Reliability

Evaluate each dimension before trusting a tool's output for scholarly work.

✓ Citation Traceability

Can every cited paper be retrieved via DOI or standard academic search? If not, the citation may be fabricated.

✓ Source Coverage

Does the tool index the journals and conferences relevant to your field? Coverage gaps produce systematic blind spots.

✓ Freshness

Is recent research indexed? LLM knowledge cutoffs may exclude work from the past 12–24 months — critical in fast-moving fields.

✓ Transparency

Does the tool disclose its data sources, indexing methodology, and known limitations? Opaque tools are harder to cite in methods sections.

✓ Reproducibility

Can the same search be repeated and produce consistent results? A core requirement for systematic reviews.

⚠ Hallucination Risk

The most serious reliability failure for research use. Does the tool generate plausible-sounding but non-existent references?

"Reliability determines whether a tool can be responsibly integrated into a scholarly workflow."

Research AI Tool Evaluation Scorecard

Use this scorecard to evaluate any AI research tool before integrating it into a systematic workflow. Score each criterion **1–5** based on **documented evidence**, not marketing claims.

Criterion	Evaluation Question	Score (1–5)
Source Quality	Are sources peer-reviewed and from indexed scholarly venues?	★★★★★
Coverage	Is the literature broad enough for systematic review in your field?	★★★★★
Citation Traceability	Can every cited paper be retrieved via DOI or standard search?	★★★★★
Freshness	Is research from the past 12–24 months indexed and retrievable?	★★★★★
Transparency	Does the tool disclose its data sources and known limitations?	★★★★★
Reproducibility	Can the same search be repeated with consistent results?	★★★★★

Choose tools based on research tasks, not popularity.

Searching Is Not the Same as Discovering

A literature review built only on keyword search is structurally incomplete. Discovery methods are required to map the full intellectual landscape of a research area.

SEARCH

A 2-step retrieval process

1. Keyword Query

Enter search terms

2. Matching Results

Papers containing those terms

Fast, efficient, and **limited to what the query explicitly names**. Returns papers that match terms — not papers that **matter**.

DISCOVERY

A 6-step relational cascade

Seed Paper

Start with an anchor study

References

Trace cited works backward

Citations

Follow forward citations

Authors

Map influential researchers

Clusters

Identify conceptual groupings

Emerging Themes

Surface adjacent research frontiers

Explores the **intellectual neighbourhood**. Surfaces work that keyword search would miss.

⚠ **What search misses:** Foundational papers using different terminology
| Emerging work not yet heavily cited | Influential authors on adjacent problems

Search retrieves. Discovery explores relationships.

A Literature Review Is Not a Search Result

Each stage is distinct. AI may accelerate certain stages — it should not collapse them into one "Generate Literature Review" button.

01

Search

Retrieve candidate papers from scholarly corpora

02

Discovery

Explore citation networks and conceptual neighbourhoods

03

Screening

Filter by relevance, quality, and methodological fit

04

Reading

Extract methodological detail from primary sources

05

Critical Evaluation

Assess validity, bias, and methodological rigour

06

Comparison

Identify convergence and contradiction across studies

07

Evidence Synthesis

Integrate findings into coherent scholarly narrative

08

Gap Identification

Surface unexplored questions from the body of evidence

09

Research Question

Requires domain expertise — cannot be delegated to AI

 **AI may accelerate these stages. It should not collapse them into one "Generate Literature Review" button.**

A Literature Review Is Not a Search Result

AI Role by Stage



● AI can accelerate:

Stages 1–3 (Search, Discovery, Screening) and initial Comparison
These are retrieval and filtering tasks where AI tools can meaningfully reduce manual effort.



● AI cannot replace:

Stage 5 – Critical Evaluation
Stage 7 – Evidence Synthesis
Stage 8 – Gap Identification
These require **disciplinary judgment** and research context that AI does not possess.

Why the Distinction Matters

Collapsing all 9 stages into one AI prompt produces a **simulacrum** of a literature review – not a scholarly one.

Each stage builds on the last. Skipping or automating critical stages undermines the **validity of the research question** itself.

One Good Paper Can Open an Entire Research Field

A single high-quality, relevant paper is the most reliable entry point into an unfamiliar research area. **The quality of your seed set determines the quality of your discovery.**



Modern research discovery is network-based, not only keyword-based.

Semantic Scholar: Intelligent Scholarly Search

A free, AI-powered scholarly search engine developed by the Allen Institute for AI. Indexes **200+ million academic papers** across all disciplines.

200+ Million Papers
Largest open scholarly index across all disciplines

Free & Open Access
No login required — available to everyone

AI-Computed Similarity
Finds conceptually related papers beyond keywords

No Login Required
Instant access to structured scholarly metadata

Search Interface

 **"Concept Drift Healthcare"** — Search Semantic Scholar

RESULT 1

Gama, J. et al. · 2014

Citations: 2,847 · **Venue:** ACM Computing Surveys

A survey on concept drift adaptation — foundational reference in the field

[View References](#) | [View Citations](#) | [Related Papers](#)

RESULT 2

Bayram, F. et al. · 2022

Citations: 341 · **Venue:** Journal of Biomedical Informatics

From concept drift to model degradation — clinical ML deployment

[View References](#) | [View Citations](#) | [Related Papers](#)

RESULT 3

Nestor, B. et al. · 2019

Citations: 198 · **Venue:** NeurIPS Clinical AI Workshop

Feature robustness under distribution shift in clinical EHR models

[View References](#) | [View Citations](#) | [Related Papers](#)

Best starting point: structured scholarly discovery.

Semantic Scholar: Intelligent Scholarly Search

A free, AI-powered scholarly search engine developed by the Allen Institute for AI.
Indexes **200+ million academic papers** across all disciplines.

200+ Million Papers

Largest open scholarly index across all disciplines

Free & Open Access

No login required – available to everyone

AI-Computed Similarity

Finds conceptually related papers beyond keywords

No Login Required

Instant access to structured scholarly metadata

How to Use Semantic Scholar Strategically

Do not stop at the first search-results page. Follow this 8-step workflow to maximize scholarly discovery value.

01

Define Research Question

"How is concept drift detected in clinical prediction systems?" — more productive than broad terms like "concept drift."

03

Identify Anchor Paper

Select a paper that **closely matches** your research question. This becomes your **first anchor** for deeper exploration.

02

Initial Search

Use **specific, domain-qualified terms**. Explore the first **20–30 results** before refining your query.

04

Explore References

What did this paper build on? References reveal the **foundational literature** underpinning the field.

How to Use Semantic Scholar Strategically

05

Explore Citations

Who cited this paper? Citations reveal how the field **responded and evolved** after publication.

07

Author Exploration

What else has this author published? **Co-authors?** **Institutional affiliations?** Follow the network of expertise.

06

Related Papers

AI-computed similarity surfaces conceptually adjacent work that keyword search alone may **miss entirely**.

08

Refined Search

Use **terminology discovered in steps 3–7** to run a more precise second search with richer results.

Citation Count $\neq$ Research Quality

Paper A

Published: 2015

Citations: 5,000

Foundational method paper in general ML

vs.

Paper B

Published: 2025

Citations: 25

Empirical study of concept drift in ICU data streams with sparse labels

Which should you read first?

IT DEPENDS ON YOUR RESEARCH QUESTION.

Age Effect

Older papers accumulate more citations naturally over time

Field Size

Large fields generate more citations overall

Relevance Gap

High-citation papers may address a different problem entirely

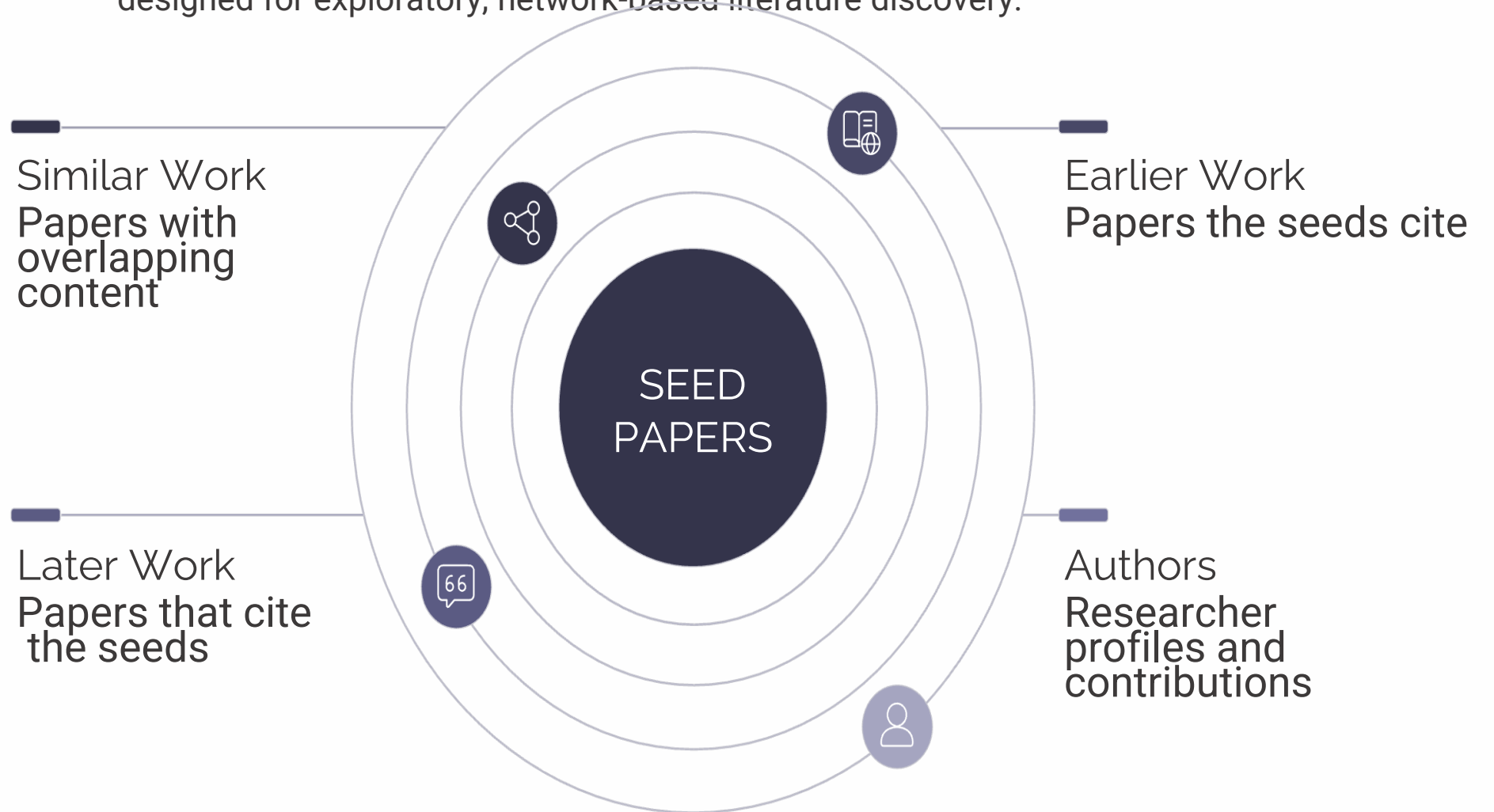
Methodological Quality

Citation count does not equal methodological rigour

Use citation count as a signal of influence — not as a proxy for relevance or quality.

Research Rabbit: Explore a Research Neighbourhood

A **free literature discovery tool** that visualizes citation relationships around a set of seed papers — designed for exploratory, network-based literature discovery.



Research Rabbit: Explore a Research Neighbourhood

A **free literature discovery tool** that visualizes citation relationships around a set of seed papers — designed for exploratory, network-based literature discovery.

Similar Work

Papers with overlapping content — surfaces work using

Earlier Work

Papers the seeds cite — intellectual origins of the research

Later Work

Papers that cite the seeds — field responses and extensions

Authors

Researcher profiles and contributions — discover key voices in the field

Collections

Curated paper sets you build, organise, and share with collaborators

 **What it is:** A free literature discovery tool that visualizes citation relationships around a set of seed papers — revealing clusters, influential nodes, and connection patterns that keyword search cannot surface.

Start with papers you trust. Expand from there.

From Seed Papers to a Literature Network



Add 3–5 Seed Papers

Select papers you have already read and trust. For concept drift in healthcare: **2–3 drift detection papers + 1–2 clinical prediction papers.**



Explore Similar Work

Surfaces papers with overlapping content — including papers that use **different terminology.**



Identify Recurring Authors

Authors appearing repeatedly across the network are likely **key contributors** to the field.



Create a Collection

Group seed papers into a named collection. **Collections are the unit of exploration** in Research Rabbit.



Follow Citation Relationships

Navigate **earlier work** (what seeds cite) and **later work** (what cites the seeds) to map the intellectual timeline.



Expand Literature Set

Add newly discovered relevant papers to the collection. **The network updates dynamically.**

The network grows as you discover.

Connected Papers: See the Structure of a Research Area

A visual tool that generates a **similarity graph** of papers related to a single seed paper. Uses a **co-citation and bibliographic coupling algorithm** — not keyword matching.

Node Size

= Citation Count

Larger nodes are more influential papers

Node Proximity

= Conceptual Similarity

Closer nodes share more conceptual overlap

Connections

= Shared Citation Relationships

Lines link papers with overlapping references

Clusters

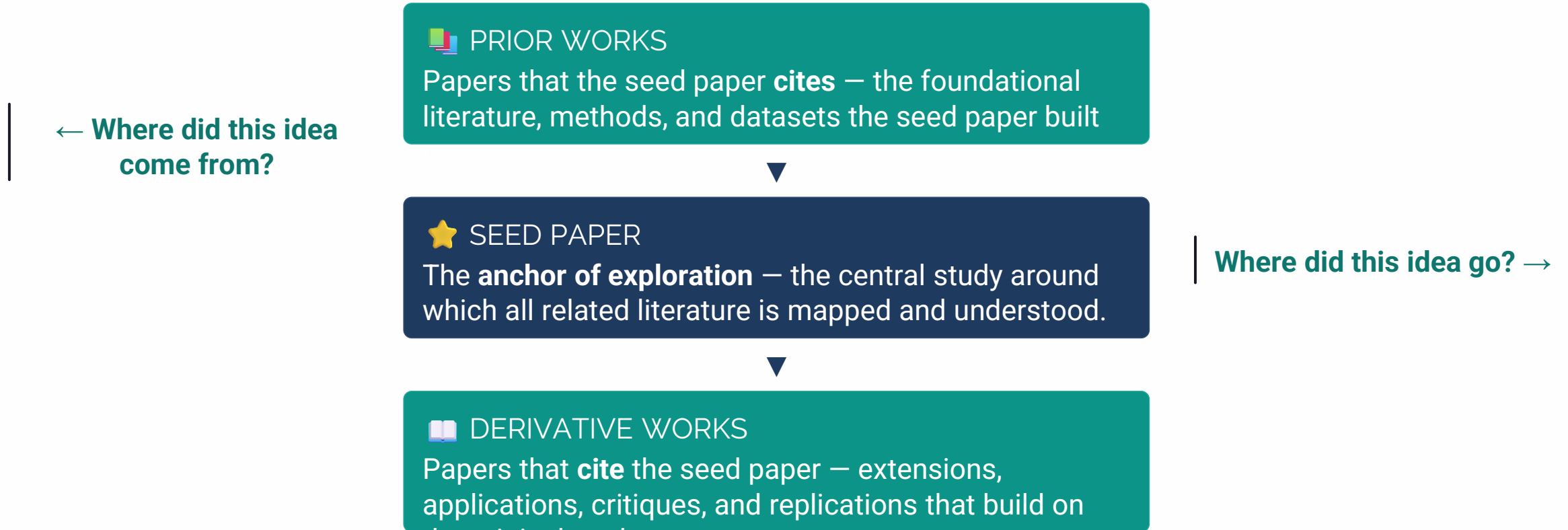
= Research Neighbourhoods

Groups reveal distinct sub-fields and themes

Best for understanding conceptual neighbourhoods and related work.

Where Did the Idea Come From, and Where Did It Go?

Understanding the **directional flow of scholarly influence** is essential for constructing a complete and rigorous literature review.



Research application — Concept Drift Detection:

Prior works: Foundational algorithms — ADWIN, DDM, EDDM, and early drift theory papers.

Derivative works: Clinical applications, streaming ML adaptations, and unsupervised drift extensions.

Research Rabbit and Connected Papers: Complementary, Not Competing

Different tools answer different research questions. Use both – at different stages of the literature review process.

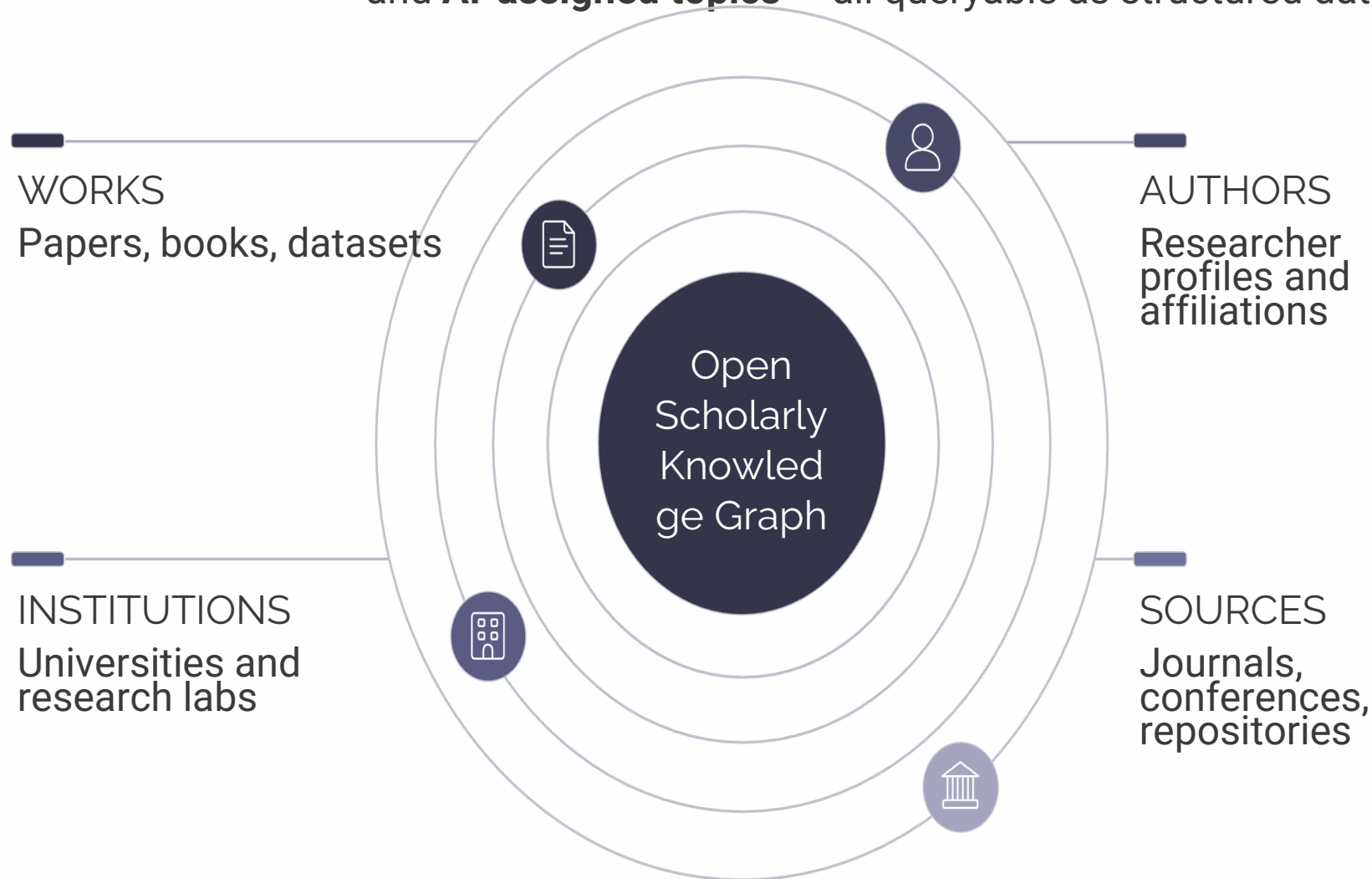
Feature	Research Rabbit	Connected Papers
Main Goal	Explore and expand literature network	Visualize conceptual relationships
Starting Point	Collection of 3–5 seed papers	Single seed paper
Author Exploration	Strong – dedicated author profiles	Limited
Citation Exploration	Strong – earlier and later work	Strong – co-citation and coupling
Ongoing Discovery	Strong – dynamic collection updates	Moderate – static per query
Best Use	Expand research network over time	Understand field structure quickly

Use **Research Rabbit** to **EXPAND** your literature network over time.

Use **Connected Papers** to **UNDERSTAND** field structure quickly.

OpenAlex: Turning Scholarly Literature Into Structured Data

A fully open, free scholarly knowledge graph connecting **250M+ works**, **90M+ authors**, **100K+ institutions**, **200K+ sources**, and **AI-assigned topics** — all queryable as structured data.



OpenAlex: Turning Scholarly Literature Into Structured Data

Research Use Cases

Publication Trend Analysis

Track publication volumes by year and topic to identify emerging research directions.

Institutional Research Output Comparison

Compare research productivity and impact across universities, labs, and funders.

Author Collaboration Network Mapping

Visualise co-authorship networks and discover key connectors in a research field.

Topic Emergence Tracking

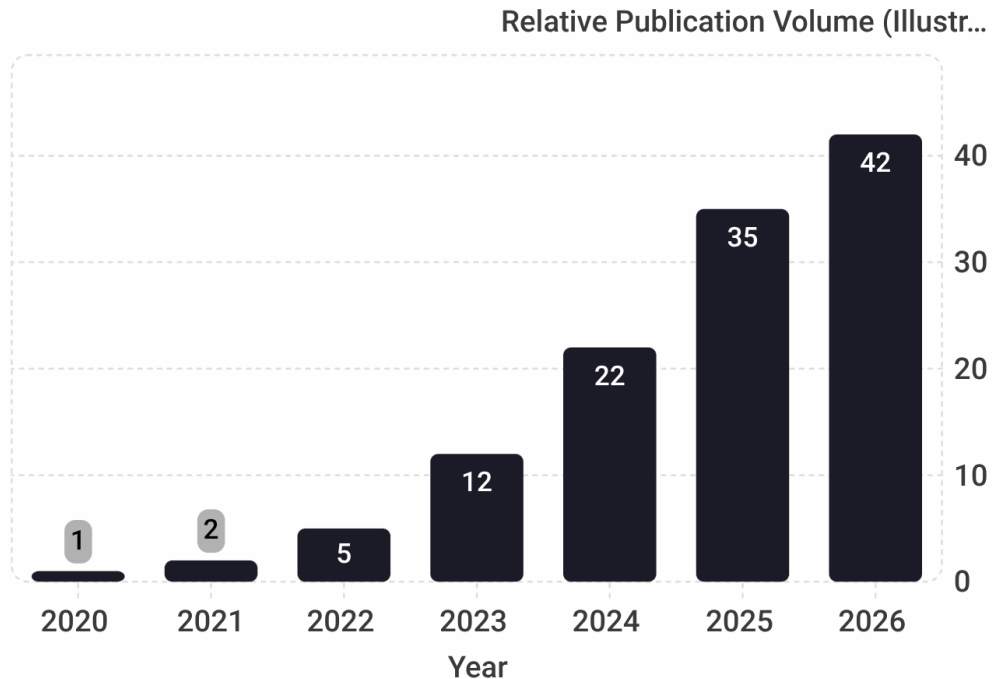
Monitor the rise and decline of research themes over time using AI-assigned concepts.

"Literature can be analysed as data, not only read as documents."

From Individual Papers to Research Trends

⚠️ **ILLUSTRATIVE TREND VISUALIZATION** — do not cite specific publication counts without verifying the data source. Publication counts vary by database, indexing lag affects recent years, and preprints may or may not be included.

LLMs in Healthcare — Relative Publication Volume (2020–2026)



Values are illustrative index scores, not verified publication counts.

Key Analytical Questions

📈 Growing, plateauing, or declining?

🏛️ Which institutions are most active?

🔍 Which subtopics are emerging?

👥 Who are the key contributors?

"Trend analysis contextualizes individual papers within the broader research trajectory."

Litmaps: Build and Maintain a Living Literature Map

A citation-mapping tool that visualizes the temporal and relational structure of a literature set. Designed for **ongoing literature monitoring** throughout a research project.



Seed Papers

Start with a set of trusted, highly relevant papers as your foundation.



Citation Map

Papers are plotted on a publication timeline with citation connections shown as linking lines.



Related Literature

Discover adjacent and emerging work — clusters and gaps become visually apparent.



Curated Collection

Build and organize your literature set with papers selected for relevance and quality.

Litmaps: Build and Maintain a Living Literature Map



New Paper Discovery

Litmaps alerts you when new relevant papers are published that match your collection.



Ongoing Monitoring

The review stays current and evolving throughout your entire PhD journey.

i Key differentiator: Litmaps is designed for **LONGITUDINAL use** — it alerts researchers when new papers are published that are relevant to their collection. Particularly valuable for PhD students conducting multi-year research.

"Literature review is a continuous process, not a one-time task."

Which Research Discovery Tool Should I Use?

Tool	Primary Strength	Best Use Case
Semantic Scholar	Scholarly search + AI similarity	Find relevant papers – starting point for any literature search
Research Rabbit	Citation network exploration	Expand literature from trusted seed papers
Connected Papers	Conceptual similarity mapping	Understand field structure and research neighbourhoods
OpenAlex	Scholarly knowledge graph + metadata	Analyse publication trends, institutions, and topic relationships
Litmaps	Citation mapping + monitoring	Maintain and monitor literature throughout a research project

There is no single best research tool.

There is a best tool for a specific research task.

How Should a PhD Student Actually Use These Tools Together?



Step 1 — Research Question

Define before opening any tool. The question determines which tools are appropriate.



Step 2 — Semantic Scholar: Discover

Structured scholarly search to find candidate papers.



Step 3 — Research Rabbit: Expand

Build a literature network from trusted seed papers.



Step 4 — Connected Papers: Map

Visualize the conceptual structure of the research area.

How Should a PhD Student Actually Use These Tools Together?



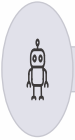
Step 5 — OpenAlex: Analyse

Examine publication trends, institutional activity, and topic relationships.



Step 6 — Litmaps: Monitor

Set up ongoing alerts for new relevant publications.



Step 7 — AI Assistant: Organize & Compare

Structure evidence and generate comparison tables from verified papers.



Step 8 — Original Papers: Verify Evidence

Read primary sources. Confirm all claims against original text. Reject unverifiable content.

Tool orchestration matters more than tool collection.

An AI-Assisted Literature Discovery Workflow

The researcher begins and ends the process. AI tools accelerate the middle stages — they do not replace the intellectual endpoints.

01

RESEARCHER — Define Research Question

The intellectual anchor. All subsequent steps serve this question. Define before opening any tool.

02

Semantic Scholar — Discover

Structured scholarly search to find candidate papers and establish first anchors.

03

Research Rabbit — Expand

Build a literature network from trusted seed papers to broaden coverage.

04

Connected Papers — Map

Visualize conceptual clusters and field structure across the research area.

05

OpenAlex — Analyse Trends

Examine publication dynamics, institutional activity, and topic emergence over time.

06

Litmaps — Monitor

Set up ongoing literature monitoring and alerts for new relevant publications.

An AI-Assisted Literature Discovery Workflow

The researcher begins and ends the process. AI tools accelerate the middle stages — they do not replace the intellectual endpoints.

07

08

09

AI Assistant — Organize & Compare

Structure evidence and generate comparison tables from verified papers only.

Verify Evidence

Read primary sources. Confirm all claims against original text. Reject unverifiable content.

Critically Synthesize

Integrate evidence, identify gaps, formulate research contribution. This step cannot be delegated. The researcher remains the sole author of intellectual synthesis.

The researcher remains the integrator of evidence.

Case Study: Concept Drift Detection in Healthcare Data Streams

Research Question: How can concept drift be detected in evolving clinical prediction systems?

01

Research Question

How can concept drift be detected in evolving clinical prediction systems?

02

Semantic Scholar — Find

Find candidate papers on drift detection in clinical and streaming healthcare contexts. Filter 2021–2026.

03

Research Rabbit — Expand

Expand the literature network from 3–5 trusted seed papers. Identify earlier foundational work and later clinical applications.

04

Connected Papers — Map

Identify conceptual clusters: drift detection methods, clinical prediction models, streaming ML, and label-sparse environments.

05

OpenAlex — Analyse

Explore publication trends. Is clinical drift detection growing? Which institutions are active?

06

Litmaps — Monitor

Set up monitoring for new papers on concept drift in healthcare. Track the field through the PhD.

Case Study: Concept Drift Detection in Healthcare Data Streams

Research Question: How can concept drift be detected in evolving clinical prediction systems?

07

Critical Reading — Compare

Compare methods, datasets, evaluation metrics, and limitations across retrieved papers.

08

Identify— Potential Gap (Illustrative)

Reliable drift detection when labels are sparse and clinical environments evolve rapidly.



⚠ Illustrative example only. The gap identified above must be validated through systematic literature analysis. Do not cite this as a confirmed research gap without independent verification.

Seven Common Mistakes Researchers Make With AI



1. Asking AI to "Give 20 References"

AI will comply — and may fabricate plausible-sounding citations that do not exist.



2. Copying AI-Generated Literature Reviews

May contain fabricated citations, misrepresented findings, and outdated information.



3. Trusting Citations Without Verification

A citation that looks correct may still be fabricated — always verify via DOI.



4. Searching Only Through Conversational AI

Not a scholarly database — generates text based on training data, not live literature.



5. Equating Citations With Quality

High citation count signals influence, not methodological quality or relevance.



6. Using One Tool for Every Task

No single tool excels at search, discovery, mapping, trend analysis, and monitoring.



7. Asking AI to Declare the Research Gap

Gap identification requires domain expertise and deep reading — it cannot be delegated.

 **Convenience can create false confidence.**

The Seven Rules of AI-Assisted Research

1

1. Search Before Synthesis

Discover and read the literature before asking AI to synthesize it.

2

2. Verify Important Citations

Every specific claim must be traceable to an original paper via DOI.

3

3. Prefer Primary Scholarly Sources

Peer-reviewed papers take precedence over summaries and AI-generated overviews.

4

4. Separate Evidence from AI Interpretation

Distinguish what a paper actually reports from what AI says it reports.

RESPONSIBLE RESEARCH

The Seven Rules of AI-Assisted Research

5

5. Use Multiple Discovery Strategies

Combine keyword search, citation network exploration, conceptual mapping, and trend analysis.

6

6. Keep Humans in the Decision Loop

Research questions, gap identification, and evidence synthesis require human judgment.

7

7. Document Meaningful AI Assistance

Transparency about AI use is an emerging ethical and methodological norm.

RESPONSIBLE RESEARCH

AI WILL NOT REPLACE RESEARCHERS.

Research Expertise × AI Literacy × Critical Verification = Responsible AI-Augmented Research

What AI Changes

- Speed of literature discovery
- Scale of information processed
- Ease of generating structured comparisons
- Accessibility of bibliometric analysis

What AI Does Not Change

- Need for domain expertise
- Requirement for critical evaluation
- Responsibility for evidence verification
- Intellectual work of identifying genuine research gaps

"The goal is not to automate research. The goal is to augment the researcher while preserving scientific rigor."

Which part of your current research workflow could AI improve tomorrow?